Deirdre Kimberley

647-924-5805 | dfxkimbe@uwaterloo.ca | linkedin.com/in/deirdrekimberley

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechanical Engineering — *Sept. 2022 – Present*

Experience

Apple — April 2026 – Present
Manufacturing Quality Engineer – Home/Audio — *Cupertino, CA*

- Led R&D and deployment of metrology systems for **previously unverifiable critical dimensions**, including material waviness, birefringence, and IR-leakage, from concept through full machines
- Owned the DFM and dimensional convergence for 5 parts through early builds, closing **160+ nonconformances** through tolerance/capability studies, CpK/FAI validation, DOE trials, and datum/GD&T scheme convergence
- Delivered two equipment programs on schedule: a DFM-driven machine retrofit and a new production line quality control plan, aligning vendors and engineering teams from concept through implementation

Tesla — Sept. 2025 – Dec. 2025
Technical Program Manager – Optimus — *Palo Alto, CA*

- Owned the end-to-end manufacturing program for the hand line, coordinating cross-functional teams and defining **120+ manufacturing interfaces** for automation equipment design
- Resolved **200+ build issues**, driving alignment across design, manufacturing, quality, and production teams to maintain build targets and ensure DFM/DFA compliance
- Delivered **34 line fixtures and 9 testers** for the next generation hand, validating early designs through rapid prototyping and reliability testing

Tesla — Jan. 2025 – April 2025
Manufacturing Quality Engineer – Cybercab — *Austin, TX*

- Developed 150+ PFMEA lines for wheel alignment, identifying **100+ manufacturing risks** and implementing controls for the unboxed alignment process
- Led the R&D and production validation of an automated vision inspection system, developing test methodology and qualifying detection performance for line deployment
- Redesigned the manufacturing traceability process, identifying and driving closure of **200+ gaps** to mitigate line-down risk ahead of launch

Magna International — April 2024 – Aug 2024
Research and Development Engineer — *Toronto, ON*

- Designed and implemented a cobot-based quality inspection station, developing rapid-deployment fixtures and operator software that reduced **setup time by 40%**
- Performed mechanical property testing and analyzed data across **2,000+ samples** to support R&D programs

Martinrea International — Sept. 2023 – Dec. 2023
Manufacturing Engineer — *Tillsonburg, ON*

- Designed and launched a consolidated manufacturing station, delivering **$225,000 in annual savings** while eliminating 450 sq. ft. of floor space, and 1 operator
- Redesigned production line stations using Design for Six Sigma, **increasing throughput by 50%**, reducing cycle time by **17 seconds**, and improving operator safety

Projects

Pick and Place Robot | *Python, Arduino, Motor Control, Sensor Integration* — Jan. 2026 – April 2026

- Designed the robot's electrical architecture, integrating motors, sensors, control hardware, and power systems to enable automated pick-and-place operation
- Developed motion-control and sequencing software, integrating sensor feedback and validating performance across mechanical and electrical subsystems

Technical Skills

Mechanical: SolidWorks, Catia, NX, Inventor, AutoCAD, FEA, GD&T, Root Cause Analysis, Tolerance Analysis
Electrical: Arduino, Raspberry Pi, I2C, Circuit Design, Sensor Integration, Motor Control, PCB Design
Software: Python, C++, Java, MATLAB, Excel